AI-BASED ONLINE PROCTORING SYSTEM

[Risk Scoring Agent]

1. Problem Overview and Statement

In an online examination environment, teachers or administrators may receive malpractice evidence as an uploaded image. This image may show suspicious activities such as mobile phone usage, another person near the student, a book or unauthorized material, or a mismatch in student identity.

The main problem is that manually checking every uploaded image, identifying the correct student, searching the database, checking previous violations, calculating the seriousness of the case, and preparing a final report takes time.
It can also lead to inconsistent judgement if every case is reviewed differently.

This project solves the problem by creating an AI-assisted Risk Scoring Agent. The system analyzes the malpractice image, identifies the student using profile photo and student ID card verification, detects the malpractice type, fetches student details from the database, checks previous violations, calculates a risk score, and generates a clear report for teacher/admin review.

The system is designed as a decision-support tool. It does not automatically punish a student. Instead, it gives a risk score, reasons, evidence summary, and recommendation so that the final decision can be taken by a human teacher or exam authority.

2. Implementation

Step 1 - Project Setup and Folder Structure

Create the project structure separating backend API, image analysis, identity verification, risk scoring, RAG, report generation, database, uploads, and configuration files. The main backend entry point is app.py, and the scoring logic is kept inside risk_scoring_agent.py.

Step 2 - Student Database Setup

Create database tables for students, student ID cards, identity verifications, malpractice events, RAG policy chunks, and reports. Store student_id, name, department, semester, profile_photo_path, and id_card_path.

Step 3 - Student Registration and Identity Data

Register each student with a profile photo and student ID card image. These images become the trusted reference for identity verification.

Step 4 - Teacher Malpractice Image Upload

Create an upload API where the teacher/admin uploads a malpractice image as evidence. The uploaded image is saved in the uploads/malpractice_images/ directory with its file path stored in the database.

Step 5 - Image Analysis

Send the uploaded malpractice image to the image analysis module. The image analysis module identifies the visible student and detects suspicious objects or activities.

Step 6 - Student Identification

Use face matching to compare the uploaded image face with the registered profile photo. If a student ID card is visible, use OCR to read the student ID and match it with the database record.

Step 7 - YOLO Object Detection

Use YOLO to detect malpractice-related objects or events such as mobile phone, book, earbud, another person, or face not visible. Store the event type and confidence score.

Step 8 - Fetch Student Details

After identifying the student_id, fetch student details from the database. This includes name, department, semester, exam information, profile photo path, and previous records.

Step 9 - Previous Violation Check

Check the malpractice_events table to count previous violations for the same student. Previous incidents are used to increase risk score and recommendation priority.

Step 10 - RAG Policy Document Preparation

Upload examination rules and malpractice policy documents. Split the documents into small chunks and store them in rag_chunks for retrieval during review.

Step 11 - Embeddings and Policy Retrieval

Generate embeddings for policy chunks using Gemini Embedding API or another embedding model. During review, retrieve relevant policy clauses such as mobile phone rules or identity mismatch rules.

Step 12 - Rule-Based Risk Scoring

Calculate the risk score using fixed scoring rules. For example, mobile phone detected = 40, another person detected = 45, ID mismatch = 60, low identity confidence = +15, and each previous violation = +10.

Step 13 - LLM-Based Evidence Synthesis

Send the student details, detected evidence, risk score, previous history, and retrieved policy clauses to Gemini LLM. The LLM creates a clear explanation while separating actual evidence from assumptions.

Step 14 - Human Review Recommendation

Generate a recommendation for teacher/admin review. The system may suggest Cleared, Warning, Re-exam, or Disqualified, but the final decision remains with the human authority.

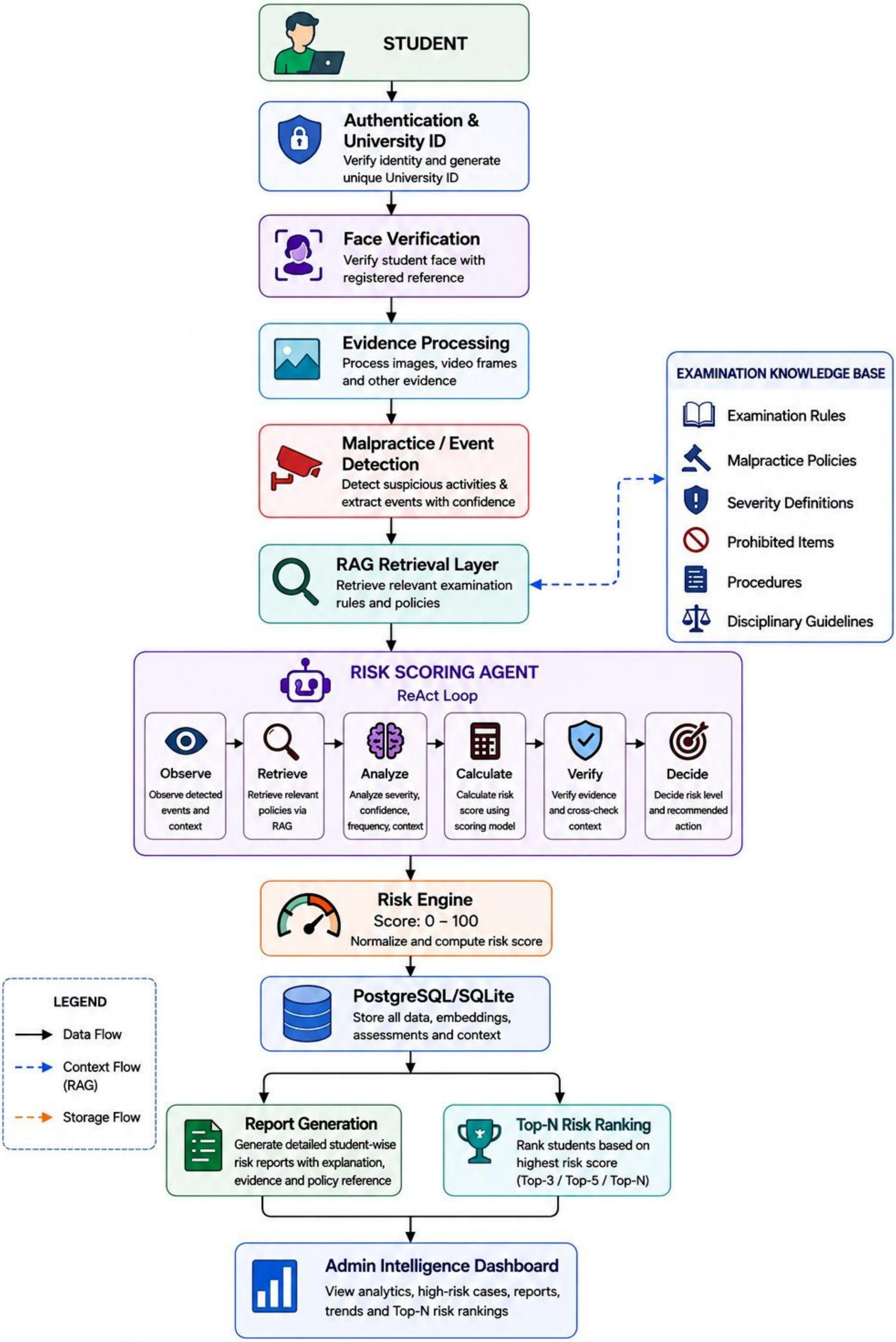
Step 15 - Final Report Generation

Generate a structured report containing student details, malpractice type, identity status, evidence confidence, previous violations, risk score, risk level, policy reference, recommendation, and admin decision.

Step 16 - Validation and Testing

Test the workflow with sample student records and sample malpractice images. Verify that image upload, student identification, object detection, database fetch, scoring, policy search, and final report generation work correctly.

3. Implementation Architecture



4. Scope of Work

- **Student Data Management:** Store student details, profile photo, and student ID card records.
- **Evidence Upload:** Allow teacher/admin to upload malpractice images for review.
- **Identity Verification:** Match uploaded image face with student profile photo and verify visible student ID card when available.
- **Computer Vision:** Detect malpractice evidence such as mobile phone, book, earbud, another person, or face not visible.
- **Database Fetching:** Retrieve student details and previous malpractice history from the database.
- **RAG-Based Policy Grounding:** Retrieve relevant exam policy rules for the detected malpractice type.
- **Risk Assessment:** Calculate risk score out of 100 and classify risk as LOW, MEDIUM, HIGH, or CRITICAL.
- **LLM Synthesis:** Generate a clear evidence-based explanation using Gemini LLM.
- **Human-in-the-Loop Review:** Provide recommendation for teacher/admin review instead of automatic punishment.
- **Final Reporting:** Produce a structured report with reasons, risk level, recommendation, and processing history.

5. Main Project Components

Component	Purpose
app.py	Starts the FastAPI backend and defines upload/review APIs.
database.py	Handles database connection, queries, and table operations.
image_analysis.py	Controls image analysis and returns student_id, malpractice type, and confidence.
identity_verification.py	Matches profile photo and checks student ID card details.
object_detection.py	Uses YOLO or a detection model to find suspicious objects/events.
risk_scoring_agent.py	Calculates risk score and risk level using scoring rules.
rag/policy_loader.py	Loads examination policy documents for RAG.
rag/vector_store.py	Stores and searches policy embeddings.
llm/gemini_client.py	Calls Gemini LLM for evidence synthesis and report explanation.
report_generator.py	Creates final text or PDF report for teacher/admin.
uploads/malpractice_images/	Stores uploaded malpractice evidence images.
student_photos/	Stores registered student profile photos.
student_id_cards/	Stores registered student ID card images.
database/proctoring.db	SQLite database for beginner version.

6. Database Tables

Table	Main Data Stored
users	Login details, role, username, email, password hash.
students	student_id, name, department, semester, profile photo path.
student_id_cards	ID card image path, extracted name, extracted student_id.
identity_verifications	Face match, ID match, confidence, overall verification status.
malpractice_events	Uploaded image path, event type, confidence, base score, status.
rag_documents	Uploaded policy document details.
rag_chunks	Policy text chunks, embeddings, embedding model name.
reports	Risk score, risk level, AI assessment, recommendation, report path.

7. Risk Scoring Logic

The first working version should use rule-based scoring because it is easy to understand, test, and explain. The score is capped at 100.

Condition	Score
Looking away	10
Face not visible	25
Mobile phone detected	40
Book detected	35
Earbud detected	35
Another person detected	45
Student ID mismatch	60
Identity confidence below 70 percent	+15
Each previous violation	+10

Score Range	Risk Level	Meaning
0 - 30	LOW	Minor suspicion. Teacher may review.
31 - 60	MEDIUM	Needs teacher verification.
61 - 80	HIGH	Strong suspicion. Investigation recommended.
81 - 100	CRITICAL	Immediate authority review recommended.

8. Expected Output

The completed system should generate a structured final report containing:

- Student ID and student name
- Department, semester, and exam ID
- Uploaded evidence image path
- Identity verification status
- Detected malpractice type
- Detection confidence score
- Previous violation count
- Relevant policy clauses from RAG
- Risk score out of 100
- Risk level
- AI-generated evidence synthesis
- Human review recommendation
- Admin decision field
- Report generation date and time

Important: The system is designed as an AI-assisted decision-support system. A high-risk result should trigger human investigation and should not, by itself, be treated as conclusive proof of cheating.

Future Scope

1. Transition from Dummy Image Analysis to Real AI Models

The beginner version may use dummy outputs for student_id and malpractice type. The next phase should replace dummy logic with real face matching, student ID card OCR, and YOLO object detection.

2. Fully AI-Assisted Risk Assessment

The future system can combine image evidence, identity status, previous history, and policy clauses. Gemini LLM can then provide contextual reasoning while rule-based scoring remains the transparent base.

3. RAG Policy Knowledge Base Improvement

More college rules, exam guidelines, and disciplinary policies can be added to improve policy retrieval and recommendation quality.

4. Bug Fixing and System Stability

Known runtime issues should be handled, including missing images, unknown student ID, low face confidence, failed OCR, model errors, invalid outputs, and database failures.

5. User Interface Development

A teacher/admin dashboard should be developed to upload evidence, view detected events, see risk score, inspect policy references, and submit final decisions.

6. End-to-End AI Proctoring Platform

The final objective is to transform the prototype into a stable AI proctoring platform where evidence is analyzed, risk is scored, reports are generated, and human review remains central.